SS-31

Barth syndrome: 40 mg SC daily; severe adult renal impairment: 20 mg daily; Community protocol: 5–40 mg/day SC

SS-31 is the synthetic aromatic–cationic tetrapeptide D-Arg–2,6-dimethyl-Tyr–Lys–Phe-NH_2, which interacts with cardiolipin-rich inner mitochondrial membranes and influences electron transport, membrane organization and oxidative stress. [5–7,18–20] For muscle strength in Barth syndrome, SS-31 (elamipretide; MTP-131) has FDA accelerated approval in adults and children weighing at least 30 kg, based on an intermediate knee-extensor-strength endpoint; verification of clinical benefit remains a condition of continued approval. [1] Broader exercise and fatigue outcomes are population-specific: the large primary mitochondrial myopathy trial was negative overall, while selected older adults showed an acute ATP-capacity signal and genotype-specific PMM analyses support a narrower exploratory benefit hypothesis. [2,4,17] Functional healthspan applications are plausible from aging-animal and mechanistic evidence. [11,19–21]

💉 💉 Tolerability supported in studied populations characterized in controlled trials and a small multiyear Barth syndrome extension; injection-site reactions are common and serious hypersensitivity is a recognized risk. [1–3,12,15]

⭐ Effectiveness is indication- and endpoint-specific: muscle-strength benefit in Barth syndrome is likely; selected bioenergetic effects are directly observed; broader physical-function and healthspan benefits are plausible. [1,2,4,7,11,17,20,21]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

Human safety observations include 24-week PMM and 48-week retinal-disease trials and a small Barth syndrome extension to 168 weeks. No serious adverse events were judged treatment-related in the two larger controlled trials, although adverse events and treatment discontinuation were more frequent with elamipretide. [2,15] The Barth extension supports sustained tolerability among patients remaining on treatment; its small size and attrition limit rare-event estimates. [12] Continued systemic tolerability under comparable monitored exposure is plausible, with separate conclusions for local, allergic, organ-specific and delayed risks below. [1,2,12,15]

Efficacy is indication-specific. MMPOWER-3 randomized 218 people with primary mitochondrial myopathy to 40 mg/day SC or placebo for 24 weeks. Neither six-minute walking distance nor total fatigue improved significantly versus placebo. This outweighs earlier exploratory signals for a broad PMM benefit, but does not prove absence of every biological effect in every population. [2]

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

Dosing & Patterns

Approved Barth syndrome regimen: 40 mg SC once daily for eligible patients weighing at least 30 kg. Adults with eGFR below 30 mL/min who are not on dialysis receive 20 mg daily; dialysis dosing and pediatric renal-impairment dosing are not established. Doses refer to active elamipretide: 40 mg corresponds to 46.8 mg elamipretide hydrochloride in the approved product. [1]

Research exposures: PROGRESS-HF tested 4 or 40 mg/day SC for 28 days without improving its primary cardiac endpoint. The older-adult study used a single two-hour IV infusion at 0.25 mg/kg/hour, about 34 mg total for a 150 lb subject, not a daily low-dose SC regimen. [4,8]

Human PK: SC bioavailability is approximately 92%, with peak concentrations at 0.5–1 hour. [1] Biological persistence varies by endpoint: MMPOWER-2 fatigue scores returned toward baseline after withdrawal; the older-adult ATP-capacity difference was absent at day 7. [3,4] In a small old-mouse withdrawal cohort, improved diastolic function persisted for two weeks after an eight-week course, declined by approximately half at four weeks, and was no longer statistically different from baseline at that time. This does not establish a universal duration of action or rule out tissue-specific repair. [11]

Speculative community dosing frequently discussed is

- **5–40 mg/day SC** — not a validated regimen for energy, recovery or healthspan.

The clinical evidence above concerns specified populations and protocols. [1–4,8]

Benefits

- **Likely improvement in muscle strength in Barth syndrome:** accelerated approval rests on knee-extensor-strength gains during open-label follow-up. The small randomized phase did not establish superiority on walking distance, fatigue or knee-extensor strength; the extension is uncontrolled. [1,12,13,16]
- **Observed acute ATP-capacity signal in selected older adults:** plausible functional benefit**:** among 39 participants selected for low mitochondrial function, ATPmax increased 27% versus 12% with placebo. Resting coupling and the prespecified muscle-fatigue analysis were negative. Post hoc analyses using contraction counts and later testing days showed exploratory fatigue-resistance signals. [4]
- **Plausible genotype-specific exercise benefit:** MMPOWER-2 had secondary fatigue signals but a nonsignificant primary walking result; MMPOWER-3 did not confirm an overall walking or fatigue benefit. [2,3] A post hoc analysis found a week-24 walking improvement in replisome-variant patients with chronic progressive external ophthalmoplegia. **Small subgroup size, selection after the overall result, and lack of parallel improvement across other endpoints limit this inference**. [17]
- **Observed reduction of oxidative stress and tissue injury in animal models:** plausible tissue protection in corresponding human settings: liver and hind-limb ischemia experiments

support effects on inflammatory macrophage polarization and autophagic flux. [9,10] Human myocardial reperfusion benefit was not confirmed: EMBRACE STEMI failed its primary infarct-size biomarker endpoint. [14]

- **Observed improvement in respiration in failing human heart tissue:** direct ex vivo exposure improved mitochondrial respiratory function, with no comparable effect in nonfailing samples. [7] Clinical heart-failure benefit remains a separate question: PROGRESS-HF was negative, and a 2026 established-HFpEF rat study improved respiration without improving diastolic function or remodeling, with modest adverse systolic and vascular changes. [8,22]
- **Plausible improvement of selected age-related functional deficits:** old-mouse studies showed improvements in diastolic function, exercise performance, ADP sensitivity, skeletal muscle force or fatigue resistance. [11,19–21] A 2025 study partly improved functional and frailty measures without detectable changes in tissue epigenetic or transcriptomic age, separating functional benefit from a claim of global biological-age reversal. [21]

Summarizing these different findings, positive findings we most likely for subjects with disease, injury, or advanced age, and were most likely absent for otherwise healthy research subjects.

Investigated Uses and Potential Applicaitons

- **Primary mitochondrial myopathy:** investigated; overall phase 3 endpoints negative, with exploratory genotype-specific signals. [2,17]
- **Barth syndrome:** approved for muscle strength in the labeled weight group. [1] Compassionate-use pediatric reports also describe cardiac and functional improvement; a 2026 five-year-old case received concurrent maximal heart-failure therapy, so attribution is uncertain and the report does not expand the approved indication. [23]
- **Heart failure and reperfusion injury:** experimental, with negative controlled human primary endpoints in the studies summarized here. [8,14]
- **Retinal disease and aging:** investigational. ReCLAIM-2 did not meet its primary low-luminance acuity or geographic-atrophy growth endpoints, but secondary ellipsoid-zone preservation and a visual-responder signal support further study; these nominal findings should not be treated as confirmed visual efficacy. [15] Aging-animal results support functional healthspan hypotheses. [11,20,21]

Cautions

- **Pregnancy, lactation and reproductive effects**: Human pregnancy and milk-exposure data are insufficient. Rat and rabbit developmental studies and rat fertility testing were reassuring under the studied conditions, supporting a preclinical inference rather than established human reproductive safety. [1]
- **Renal and hepatic considerations**: Renal impairment increases exposure and requires the labeled dose adjustment in adults with severe impairment; dialysis dosing remains unresolved. Hepatic metabolism is not expected to be a major elimination pathway. These PK findings do not independently establish long-term renal or hepatic safety; monitored trial

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

experience supports plausible continued organ tolerability at comparable exposure. [1,2,12,15]

- **Cardiovascular and other longer-term risks**: No clinically relevant QTc prolongation was identified at three times the maximum recommended-dose peak concentration. [1] Stable HFrEF participants had similar adverse-event rates over 28 days, but unstable cardiac illness was not the studied population and is not itself a labeled contraindication. [8] These observations support plausible electrical and short-term cardiovascular tolerability, not prevention of disease progression. Genotoxicity tests were negative; carcinogenicity studies have not been conducted, so long-latency cancer risk cannot be confidently graded. Neurological, endocrine and metabolic endpoints were not studied comprehensively enough to assign a strong domain-specific multiyear safety inference. [1,2,12,15]

Side effects

- **Allergic reactions:** serious hypersensitivity may appear within minutes or after months of treatment. Prior serious hypersensitivity to elamipretide or an excipient is a contraindication; the label directs discontinuation and no rechallenge after a serious reaction. [1]
- **Eosinophil increases** — often emerged after at least 30 days, peaked around 90 days and returned toward baseline over 6–12 months with continued treatment or after discontinuation, generally without associated clinical manifestations in the label data. (Eosinophils are specialized white blood cells involved in allergic responses, parasite defense, immune signaling, and tissue repair. An increased eosinophil count can indicate greater immune activity, but it does not necessarily mean improved immune function.) [1]
- **Injection-site reactions** are the main common tolerability burden: they occurred in all 12 elamipretide-treated Barth trial participants. [1]

Diminished Response and Monitoring

SS-31 acts through mitochondrial membrane and protein interactions rather than a conventional cell-surface receptor agonist pathway, so classic receptor-desensitization models are not applicable. This does not exclude adaptation of mitochondrial pathways or immune responses. [5,6,18–20]

Biological mechanism

Cardiolipin helps organize respiratory-chain proteins and mitochondrial cristae. SS-31 interacts with cardiolipin-containing membranes and the cardiolipin–cytochrome c complex; experiments show reduced peroxidase activity and improved electron transfer and ATP synthesis. [5,6] Biophysical experiments indicate that membrane surface charge influences peptide binding and that SS-31 modulates surface electrostatics and lipid packing without disrupting lamellar bilayers. [18] Aged-cardiomyocyte experiments support interaction with ANT1 and the ATP synthasome, reduced excess proton leak and permeability-transition activity; related aging studies link improved ANT-mediated ADP uptake and sensitivity with functional effects. [19,20] These mechanisms plausibly reduce oxidative injury and improve cellular resilience in

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

dysfunctional mitochondria. Ex vivo failing-heart respiration improved, whereas resting ATP-to-oxygen coupling did not improve in the older-adult infusion trial. [4,7]

Conclusions

SS-31 has a defined muscle-strength indication in Barth syndrome under accelerated approval. The uncontrolled multiyear extension supports continued investigation, while the negative randomized endpoints and population-specific findings remain central to interpretation. [1,12,13,16] Overall PMM, HFrEF, myocardial reperfusion and retinal primary endpoints were negative in the cited controlled trials; genotype-specific, structural and mechanistic signals warrant their own qualified interpretation. [2,8,14,15,17]

The older-adult study supports an acute bioenergetic signal with exploratory later fatigue-resistance findings, not a demonstrated subjective energy boost. [4] Animal functional improvements and mitochondrial mechanisms make selected healthspan applications plausible; functional improvement need not imply a measurable reversal of biological age. [11,18–21]

Durability should be assessed by tissue, endpoint and treatment course: transient human ATP and fatigue trajectories and partly persistent old-mouse cardiac effects do not resolve long-term repair in healthy people. [3,4,11]

References

[1] US Food and Drug Administration. Forzinity (elamipretide) injection: prescribing information. Revised September 2025. Accessed October 27, 2026. Source

[2] Karaa A, Bertini E, Carelli V, et al. Efficacy and Safety of Elamipretide in Individuals With Primary Mitochondrial Myopathy: The MMPOWER-3 Randomized Clinical Trial. *Neurology*. 2023;101(3):e238-e252. doi:10.1212/WNL.0000000000207402

[3] Karaa A, Haas R, Goldstein A, Vockley J, Cohen BH. A randomized crossover trial of elamipretide in adults with primary mitochondrial myopathy. *J Cachexia Sarcopenia Muscle*. 2020;11(4):909-918. doi:10.1002/jcsm.12559

[4] Roshanravan B, Liu SZ, Ali AS, et al. In vivo mitochondrial ATP production is improved in older adult skeletal muscle after a single dose of elamipretide in a randomized trial. *PLoS One*. 2021;16(7):e0253849. doi:10.1371/journal.pone.0253849

[5] Szeto HH. First-in-class cardiolipin-protective compound as a therapeutic agent to restore mitochondrial bioenergetics. *Br J Pharmacol*. 2014;171(8):2029-2050. doi:10.1111/bph.12461

[6] Birk AV, Chao WM, Bracken C, Warren JD, Szeto HH. Targeting mitochondrial cardiolipin and the cytochrome c/cardiolipin complex to promote electron transport and optimize mitochondrial ATP synthesis. *Br J Pharmacol*. 2014;171(8):2017-2028. doi:10.1111/bph.12468

[7] Chatfield KC, Sparagna GC, Chau S, et al. Elamipretide Improves Mitochondrial Function in the Failing Human Heart. *JACC Basic Transl Sci*. 2019;4(2):147-157. doi:10.1016/j.jacbts.2018.12.005

[8] Butler J, Khan MS, Anker SD, et al. Effects of Elamipretide on Left Ventricular Function in Patients With Heart Failure With Reduced Ejection Fraction: The PROGRESS-HF Phase 2 Trial. *J Card Fail*. 2020;26(5):429-437. doi:10.1016/j.cardfail.2020.02.001

[9] Shang L, Ren H, Wang S, et al. SS-31 Protects Liver from Ischemia-Reperfusion Injury via Modulating Macrophage Polarization. *Oxid Med Cell Longev*. 2021;2021:6662156. doi:10.1155/2021/6662156

[10] Yang Q, Li C, Chen Q. SS31 Ameliorates Oxidative Stress via the Restoration of Autophagic Flux to Protect Aged Mice From Hind Limb Ischemia. *Front Cardiovasc Med*. 2022;9:789331. doi:10.3389/fcvm.2022.789331

[11] Chiao YA, Zhang H, Sweetwyne M, et al. Late-life restoration of mitochondrial function reverses cardiac dysfunction in old mice. *Elife*. 2020;9:e55513. doi:10.7554/eLife.55513

[12] Thompson WR, Manuel R, Abbruscato A, et al. Long-term efficacy and safety of elamipretide in patients with Barth syndrome: 168-week open-label extension results of TAZPOWER. *Genet Med*. 2024;26(7):101138. doi:10.1016/j.gim.2024.101138

[13] ClinicalTrials.gov. A Trial to Evaluate Safety, Tolerability and Efficacy of Elamipretide in Subjects With Barth Syndrome (TAZPOWER). NCT03098797. Updated April 16, 2024. Accessed October 27, 2026. Source

[14] Gibson CM, Giugliano RP, Kloner RA, et al. EMBRACE STEMI study: a Phase 2a trial to evaluate the safety, tolerability, and efficacy of intravenous MTP-131 on reperfusion injury in patients undergoing primary percutaneous coronary intervention. *Eur Heart J*. 2016;37(16):1296-1303. doi:10.1093/eurheartj/ehv597

[15] Ehlers JP, Hu A, Boyer D, et al. ReCLAIM-2: A Randomized Phase II Clinical Trial Evaluating Elamipretide in Age-related Macular Degeneration, Geographic Atrophy Growth, Visual Function, and Ellipsoid Zone Preservation. *Ophthalmol Sci*. 2025;5(1):100628. doi:10.1016/j.xops.2024.100628

[16] Thompson WR, Hornby B, Manuel R, et al. A phase 2/3 randomized clinical trial followed by an open-label extension to evaluate the effectiveness of elamipretide in Barth syndrome, a genetic disorder of mitochondrial cardiolipin metabolism. *Genet Med*. 2021;23(3):471-478. doi:10.1038/s41436-020-01006-8

[17] Karaa A, Bertini E, Carelli V, et al. Genotype-specific effects of elamipretide in patients with primary mitochondrial myopathy: a post hoc analysis of the MMPOWER-3 trial. *Orphanet J Rare Dis*. 2024;19(1):431. doi:10.1186/s13023-024-03421-5

[18] Mitchell W, Ng EA, Tamucci JD, et al. The mitochondria-targeted peptide SS-31 binds lipid bilayers and modulates surface electrostatics as a key component of its mechanism of action. *J Biol Chem*. 2020;295(21):7452-7469. doi:10.1074/jbc.RA119.012094

[19] Zhang H, Alder NN, Wang W, Szeto H, Marcinek DJ, Rabinovitch PS. Reduction of elevated proton leak rejuvenates mitochondria in the aged cardiomyocyte. *Elife*. 2020;9:e60827. doi:10.7554/eLife.60827

[20] Pharaoh G, Kamat V, Kannan S, et al. The mitochondrially targeted peptide elamipretide (SS-31) improves ADP sensitivity in aged mitochondria by increasing uptake through the adenine nucleotide translocator (ANT). *Geroscience*. 2023;45(6):3529-3548. doi:10.1007/s11357-023-00861-y

[21] Mitchell W, Pharaoh G, Tyshkovskiy A, Campbell M, Marcinek DJ, Gladyshev VN. The Mitochondria-Targeted Peptide Therapeutic Elamipretide Improves Cardiac and Skeletal Muscle Function During Aging Without Detectable Changes in Tissue Epigenetic or Transcriptomic Age. *Aging Cell*. 2025;24(6):e70026. doi:10.1111/acel.70026

[22] Schauer A, Jahn D, Vahle B, et al. Mitochondrial Targeting by Elamipretide Improves Myocardial Bioenergetics Without Translating into Functional Benefits in HFpEF. *Int J Mol Sci*. 2026;27(2):1060. doi:10.3390/ijms27021060

[23] Amedro P, Andrianoely M, Gohier P, et al. Elamipretide in pediatric Barth syndrome: from heart failure to school return. *Orphanet J Rare Dis*. 2026;21(1):340. doi:10.1186/s13023-026-04487-z